

Human Internal Exposures of Bisphenol A and Six Data-Poor Analogs Predicted by Physiologically Based Kinetic Models with Multimodal Parametrization

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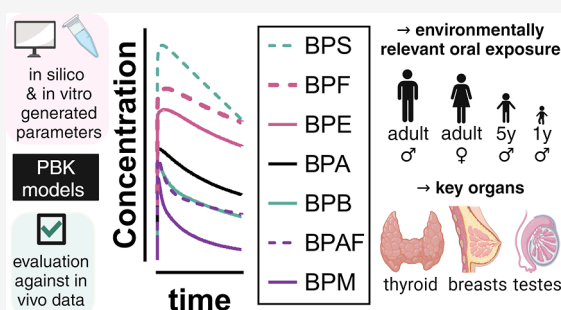
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Supporting Information

ABSTRACT: Bisphenols (BP) AF, B, E, F, M, and S are increasingly used as bisphenol A (BPA) substitutes. Despite widespread exposure and potential adverse health outcomes, they are poorly understood in terms of toxicokinetics, i.e., their absorption, distribution, metabolism, and excretion. We thus developed physiologically based kinetic models for different human physiological standards to predict internal concentrations of prevalent bisphenols following oral exposure. To address the imbalances in available human data among these chemicals, we used multimodal parametrization methods, including *in vitro* measurements of metabolism, computational prediction of gastrointestinal absorption, and rat–human extrapolation of enterohepatic circulation. Then, the models were evaluated against available human toxicokinetic data for BPA and BPS, revealing that 66% of predicted $C_{\max}$, $t_{\max}$, and AUC values fell within a 2-fold difference from *in vivo* measures. Using environmentally relevant exposure levels to compare internal levels of all tested bisphenols, we observed significant differences in the toxicokinetic profiles. Concerning tissues of toxicological concern, BPS had the highest concentration in blood and testes, while BPM accumulated in the thyroid and BPAF in the breasts. The present models are expected to facilitate a more precise evaluation of health risks induced by BPA analogs, guiding their safer use.

KEYWORDS: bisphenols, BPA, PBK modeling, ADME, pharmacokinetics, endocrine disrupting chemicals



INTRODUCTION

The rigid molecular structure of bisphenol compounds confers essential material properties for plastic production, such as chemical resistance and thermal stability.¹ In numerous countries,^{2–6} bisphenol A (BPA), once an industry staple, now faces restrictions or bans, leading to an increasing use of its structural analogs as substitutes. BPAF, BPB, BPE, BPF, BPM, and BPS were detected in beverage packaging materials.⁷ Bisphenol substitutes can migrate from packaging materials to their content,⁸ which raises significant concerns, as their structural similarity to BPA suggests they might also exert endocrine disrupting (ED) effects by binding to biological receptors.^{6,9} These interactions can lead to adverse outcomes, notably, reproductive disorders, thyroid and gonadal dysfunction/cancer, breast cancer, obesity, and diabetes.¹⁰ BPB and BPS have been identified as substances of very high concern under the European REACH Regulation due to their ED properties and reproductive toxicity.^{11,12} BPAF meets the ED criteria for human health¹³ and the opinion of ECHA's committee for risk assessment on BPAF as toxic to reproduction (Repr. 1B) is in the European Commission for decision-making.¹⁴

To evaluate the safety of BPAF, BPB, BPE, BPF, BPM, and BPS, Next Generation Risk Assessment (NGRA) principles

recommend an exposure-led approach.¹⁵ This involves converting external exposure data into concentration–time profiles in blood and organs, underscoring the necessity of prior knowledge of absorption, distribution, metabolism, and excretion (ADME) processes to determine relevant points of departure for hazard predictions. Human biomonitoring data confirms the widespread exposure to BPA analogs resulting in important human exposure, particularly among vulnerable populations, yet fails to provide essential toxicokinetic insights (Table S1).^{16–25}

The kinetic behavior of BPA following oral administration has been well characterized.^{26,27} After a rapid and efficient absorption in the gastrointestinal tract, its phase II conjugation in the gut wall and the liver leads to the extensive formation of its main metabolite, BPA glucuronide, which is rapidly eliminated via urine.²⁶ To predict chemical concentrations in

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Table 1. Exposure Scenarios, Scope, and Purpose of Modeling

scenario	purpose	model	BP	dosing	key organ(s)
1	EHC determination	rat	BPAF	single dose ^a : 340 mg/kg bw	blood
2	EHC determination	rat	BPF	single dose ^a : 200 mg/kg bw	blood
3	model evaluation	man ^b	BPA	single dose ^a : 30 μg/kg bw	blood
4	model evaluation	adult ^c	BPS	single dose ^a : 8.1 μg/kg bw	blood
5	internal exposure assessment	man	^d	single dose ^a : 336 ng/kg bw	blood, thyroid, and testes
6	internal exposure assessment	woman	^d	single dose ^a : 389 ng/kg bw	blood, thyroid, and breasts
7	internal exposure assessment	child	^d	single dose ^a : 818 ng/kg bw	blood, thyroid, and testes
8	internal exposure assessment	toddler	^d	single dose ^a : 869 ng/kg bw	blood, thyroid, and testes
9	internal exposure assessment	man	^d	repeated dose ^e : 112 ng/kg bw	blood, thyroid, and testes
10	internal exposure assessment	woman	^d	repeated dose ^e : 130 ng/kg bw	blood, thyroid, and breasts
11	internal exposure assessment	child	^d	repeated dose ^e : 273 ng/kg bw	blood, thyroid, and testes
12	internal exposure assessment	toddler	^d	repeated dose ^e : 290 ng/kg bw	blood, thyroid, and testes

^aSingle oral dosing at $t = 0$ h. Total duration of the simulation: 48 h. ^bBW and H adjusted to the population (10 men) of the toxicokinetic study by Teeguarden et al.³⁰ ^cBW and H adjusted to the population (4 men and 3 women) of the toxicokinetic study by Oh et al.³¹ ^dBPA, BPAF, BPB, BPE, BPF, BPM, and BPS. ^eRepeated oral dosing at $t = 0, 6, 12, 24, 30, 36, 48, 56, 60, 72, 78,$ and 84 h. Total duration of the simulation: 96 h.

key organs, further mechanistic insights were provided by physiologically based kinetic (PBK) models. An eight-compartment PBK model capable of predicting human internal BPA exposures was proposed by Yang et al.²⁸ It is sometimes assumed that other bisphenols undergo similar ADME patterns,²⁹ but since even slight differences in xenobiotic structures can induce significant variations in ADME processes, this application of read-across for their risk assessment should be evaluated. Of note, while the toxicokinetics of BPA³⁰ and BPS³¹ have been studied, the fate of BPAF, BPB, BPE, BPF, and BPM was not assessed in humans. This lack of data limits the applicability of the PBK model parametrization method of Yang et al.²⁸ to other bisphenols.

Karrer et al.³² addressed the challenge of extending the existing BPA PBK model to BPS, BPAF, and BPF. Their approach relied on molecular weight (MW) and structural similarities: BPA and BPF were grouped together, as were BPS and BPAF, to allow the reuse of toxicokinetic measured data from well-characterized bisphenols for calibrating models of data-poor analogs. Although this model extended predictions to less-characterized bisphenol substitutes, its application remains limited to bisphenols with minimal physicochemical differences to BPA and BPS.

To address the lack of human toxicokinetic data for a wider diversity of bisphenols, we constructed PBK models for BPA and six of its structural analogs using the OECD methodology (Figure S1).³³ We hypothesized that these models could predict internal exposures following oral administration, achieving a level of accuracy comparable to that of previous models developed with calibration to human data.^{31,32} A common concept was established for four standard human models varying by age and sex. The models for each bisphenol were individually parametrized using a multimodal approach, which integrates quantitative structure–activity relationship (QSAR) methods, quantitative in vitro to in vivo extrapolation (QIVIVE), glucuronidation kinetics measured in S9 fractions, and rat PBK modeling. The model performance was evaluated against the human toxicokinetic data. Concentrations in 12 compartments including blood, thyroid, testes, and breasts were predicted after environmentally relevant oral exposure of BPA, BPAF, BPB, BPE, BPF, BPM, and BPS.

MATERIALS AND METHODS

Scope and Purpose of Exposure Scenarios. Diet is the main route of human exposure to bisphenols; therefore, models were designed herein for oral administration.³⁴ Several exposure scenarios were defined for parameter gathering, model evaluation, and internal exposure assessment (Table 1). In scenarios 1 and 2, a rat PBK model was used to parametrize enterohepatic circulation (EHC) rates for application in human PBK models. Scenarios 3 and 4 replicated the administration of BPA from Teeguarden et al.³⁰ and BPS from Oh et al.³¹ for model evaluation. To illustrate potential BPA replacement by BPAF, BPB, BPE, BPF, BPM, and BPS, scenarios 5 to 12 assumed consistent exposure rates among all these bisphenols, based on the 2015 EFSA assessment of BPA external exposure.³⁵ Models were developed for four human physiologies: male (“man”) and female adults (“woman”), male “child”, and male “toddler” (Table S2). These models were selected based on available physiological data for key organs (thyroid, breasts, and testes), related to ED and the detected bisphenol exposure level. The repeated dose scheme from Karrer et al.³² was reproduced (scenarios 9 to 12) to model chronic exposure via three meals a day.

A common model structure was employed for the seven bisphenols (Figure 1). When not individually compartmentalized, organs were categorized as either slowly perfused tissues or rapidly perfused tissues and combined. Compartments related to the thyroid, testes, and breasts (depending on sex) were included in human but not rat models. Code was compiled in Berkeley Madonna 10 and Python (see Supporting Information and GitLab: https://gitlab.ethz.ch/eth_toxlab/pbk_model_bisphenol_analogs).

Multimodal Parametrization of PBK Models. These PBK models were parametrized using a multimodal framework that integrates new approach methodology tools into a reproducible chemical-family wide workflow. This framework combines QSARs, in vitro experiments, and cross-species extrapolation to parametrize PBK models for a family of compounds, some of which were considered data-poor (Figure 2). These methods were chosen for their simplicity of use, with QSARs, for example, requiring minimal prior information such as molecular structures, and for their compatibility with an iterative verification process in which predictions were compared to previously available results to ensure the biological accuracy of the parameters. By harmonizing the

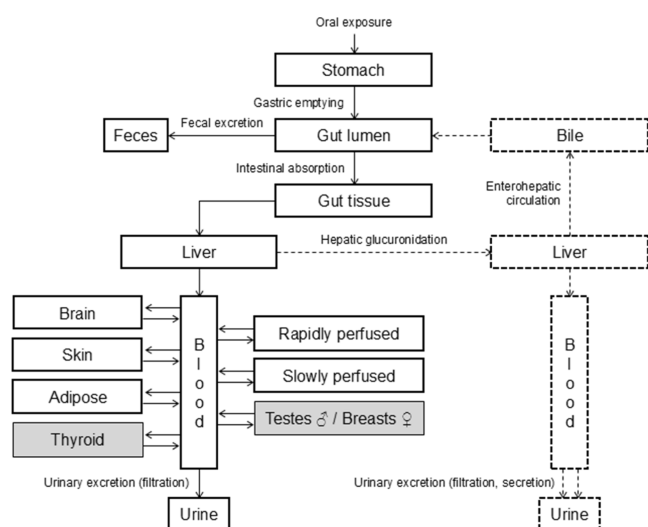


Figure 1. Schematic PBK model concept for BPA analogs (flows and compartments with continuous lines) and their glucuronidated metabolites (flows and compartments with dashed lines). Compartments with gray filling are only present in human models.

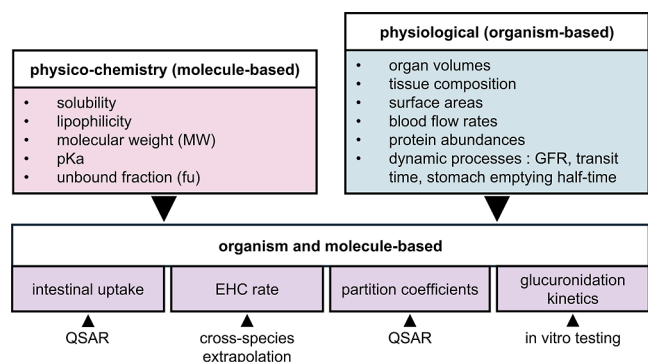


Figure 2. Overview of parameters used in the present model and how they were acquired.

parametrization process in an *in vivo*-data-independent manner across an entire compound class, this multimodal framework ensures methodological consistency, reproducibility, and applicability to other families of compounds. Parameters related to physicochemical characteristics of seven bisphenols and their glucuronides were set to the mean of values predicted from their chemical structures with ChemDraw 20.0 and Chemaxon Playground v1.6.2 (Figure 2 and Tables S3 and S4). Organism-based parameters were tailored to account for species, sex, and age when possible (Figure 2 and Table S2).^{36–49} Parameters based on both the organism and chemical were partition coefficients, intestinal uptake, EHC rate, and glucuronidation kinetics (Figure 2).

When possible, partition coefficients and plasma unbound fractions were calculated with the QIVIVE toolbox,⁵⁰ using the methods of Rodgers and Rowland⁵¹ and of Lobell and Sivarajah,⁵² and used for different ages, sex, and species (Supporting Information Table S5). For combined compartments, weighted sums of partition coefficients for the constituent organs were used. Partition coefficients of thyroid and testes were calculated using data from Pilari et al.⁵³ Based on Ulaszewska et al.,⁵⁴ breasts partition coefficients were calculated as the product of adipose partition coefficients and fat volume in breasts (24.5%).⁵⁵

To parametrize the intestinal uptake (k_a) of all bisphenols, Caco-2 apparent permeability coefficients (P_{app}) were only available for BPA, BPF, and BPS,⁵⁶ so two QSARs were additionally used to predict P_{app} of the seven bisphenols, here referred to as “QSAR K” (Kamiya et al.⁵⁶) and “QSAR L” (Lanevskij and Didziapetris).⁵⁷ LogD (lipophilicity) values were computed using the Chemaxon predictor,⁵⁸ considering pH values of 6.0 (apical) and 7.4 (basal). For “QSAR L”, standard values were set for pH (7.4), stirring rate (0 rpm), and average pore radius (5 Å), and McGowan characteristic volumes were calculated according to Zhao et al.⁵⁹ Then, all available P_{app} values per bisphenol were averaged. The scaling algorithm of Sun et al.⁶⁰ was applied (“all drugs” regression; pH 7) to obtain the *in vivo* effective permeability (P_{eff}) per intestinal surface area.

Glucuronidated bisphenols were assumed to undergo EHC and renal secretion, both relying on active transporters present in the liver and the kidney. We modeled these processes using a common parameter, “EHC_r” (EHC rate), to describe them. We modeled EHC by simulating hepatic glucuronidated bisphenol cycling through bile into the gut lumen, where it reverts to its parent form (Figure 1). In this context, EHC_r represents the portion of metabolite leaving the liver to undergo biliary excretion. Then, bile transfer from the bile compartment to the gut lumen is assumed to be immediate and complete (Figure 1). Renal secretion was represented from the blood compartment, factorized by the blood flow rate to the kidneys, EHC_r representing here the transport via efflux pumps, scaled to the renal context by the ratio of volume between the kidneys and the liver, assuming comparable expression levels.⁶¹ The value of the EHC rate was constrained between 0 and 1 and fitted as described.⁶² Predicted-to-observed C_{max} ratios in blood were calculated by adjusting EHC_r until rat blood concentrations matched values from animal studies for BPF⁶³ or BPA⁴⁸ at 24 and 48 h, respectively, while maintaining the C_{max} ratio. EHC_r values for other bisphenols were calibrated by MW,⁶⁴ with BPAF as the lightest and BPF the second heaviest. For simplicity, the EHC_r of BPM was set equal to BPAF. Human models of the same sex used the same values, assuming stable biliary export protein expression.⁶⁵

Measurement and Parametrization of Glucuronidation Kinetics. The glucuronidation kinetics of BPA, BPB, BPE, BPF, BPAF, and BPM were determined as described in the Supporting Information (Tables S6 and S7). Outliers were excluded using the Nalimov test.⁶⁶ Glucuronidation rates were calculated from compound loss over time using GraphPad Prism 10.2.3 (2024), with Michaelis–Menten and substrate inhibition fitting algorithms applied, selecting the best fit based on maximal R^2 .

Rates were scaled by *in vitro* context: human hepatic S9 fractions with 107.3 mg/g liver,⁶⁷ rat hepatic S9 with 143 mg/g liver,⁶⁸ and human hepatic microsomes with 32 mg/g liver.^{32,69} To account for age-related UDP glucuronosyltransferase (UGT) 2B15 protein expression, ontogeny scaling established a scaling factor for glucuronidation (SF_g) kinetics (eq 1).⁷⁰ Protein abundances were based on Bhatt et al.,⁷¹ with both child and toddler models classified under early childhood.

$$SE_g = \frac{\text{protein abundance in age category of interest} \left(\frac{\text{pmol}}{\text{microsomal protein}} \right)}{\text{protein abundance in adult} \left(\frac{\text{pmol}}{\text{microsomal protein}} \right)} \quad (1)$$

Model Evaluation. The performance of rat and human PBK models was evaluated by comparing predicted blood $C_{\max}$, $t_{\max}$, and AUC values to their in vivo measured values, for BPAF,⁴⁸ BPF,⁶³ BPA,³⁰ and BPS.³¹ Toxicokinetic parameters predicted for BPA, BPA glucuronide, BPS, and BPS glucuronide were also compared to previous modeling results.^{31,32}

Additionally, to evaluate the importance of excretion routes in the model, mass balance analysis was implemented within the model code. This analysis quantified the fractions of the dose excreted via urine and feces, at 12, 24, and 48 h following a single-dose exposure. The resulting excretion fractions were then compared with reported experimental data.^{30,72} To evaluate the balance between renal and biliary elimination pathways in the PBK models, clearance values were derived from the outputs of the male model exposed to BPA (exposure scenario 5, Table 1). Total renal clearance (CL_R) was defined as the sum of parent BPA renal clearance (resulting from filtration) and BPA glucuronide renal clearance (including filtration and secretion). These clearances were calculated as the ratio between the blood AUC of either BPA or BPA glucuronide over 24 h and the corresponding cumulative urinary excretion within the same time frame. Biliary clearance (CL_{bile}) was estimated as hepatic blood flow (QL) multiplied by EHC_r. According to Derendorf and Schmidt,⁷³ the bile-to-plasma concentration ratio was derived as CL_{bile} divided by daily bile flow, set to 700 mL/day.⁴⁰ To further assess the impact of EHC on systemic exposure, the apparent fraction absorbed by BPA was estimated as the ratio of the product of total clearance (renal plus biliary) and blood AUC to the oral dose, accounting for renal and biliary elimination pathways.

Sensitivity Analysis. The impact of parameter deviation on model predictions was assessed by local and global sensitivity analysis (GSA). The local sensitivity analysis (LSA) was performed in human models for all bisphenols, based on the method by Evans and Andersen⁷⁴ as described by Aichinger et al.⁶² A parameter with an absolute sensitivity coefficient > 0.1 was considered sensitive. The GSA was performed in all human models, according to the workflow of McNally et al.,⁷⁵ consisting in the Morris screening exercise on all parameters, followed by the extended Fourier amplitude sensitivity test (eFAST) on a selected subset of parameters. To investigate parameter sensitivity in the distribution and elimination phase, the GSA was conducted on AUC between the times 0 and 4 h of simulation, using the package SALib.⁷⁶ Parameter ranges were set to cover approximately ± 2 standard deviations from the initial value, based on the coefficient of variation estimated from the literature (Table S8).^{44,49,71,77–81}

Uncertainty Analysis. To address parameter uncertainty in concentrations that were predicted with human PBK models, 10,000 Monte Carlo (MC) simulations were performed. Each sensitive parameter was resampled according to probability distributions from the literature (Table S8).^{44,49,71,77–81} To ensure physiological plausibility, EHC rates were constrained between 0 and 1, and the sum of all relative organ weights and the sum of all relative organ blood

flows were kept constant. MC simulation outcomes were analyzed by comparing the first quartile, the median, and the third quartile values of concentrations per time point.

RESULTS AND DISCUSSION

PBK Model Parametrization without In Vivo Data.

The parametrization of intestinal uptake could be achieved by combining various techniques. We compared P_{app} values of the BPA analogs, either measured in Caco-2 cells monolayers (BPA, BPF, and BPS) and/or predicted by two QSARs (all analogs) in Figure S2.^{56,57} Predictions were consistent between the applied QSARs for BPA, BPAF, BPB, BPE, and BPF, ranking the permeability in the same order, with values differing by a maximum of 11×10^{-6} cm/s. Moreover, computed and in vitro measured transport values were in line for BPA and BPF. This was not observed, however, for BPM and BPS. For BPM, QSAR L predicted lower permeability than that predicted with QSAR K. This could be attributed to differing sensitivities of the QSARs to the comparably larger molecular volume, which is considered in QSAR L. For BPS, the two QSAR-predicted P_{app} values aligned closely but were lower than reported in vitro permeability. Furthermore, the inclusion of three bisphenol compounds in the training data set of QSAR K⁵⁶ did not significantly enhance the predictions of P_{app} values compared to QSAR L,⁵⁷ indicating specialized training sets may not necessarily improve accuracy. These findings highlight the feasibility of using QSARs for modeling intestinal uptake without use of error-prone read-across while also demonstrating the value of integrating multiple methods and in vitro measurements to reveal potential biases in in silico approaches.

To develop a common model for all bisphenols using molecule-specific parameters, the parametrization of hepatic metabolism was based on glucuronidation kinetics measured in subcellular liver fractions. Glucuronidation kinetics were de novo measured in human liver S9 fractions for BPA, BPB, BPE, and BPM, while for BPS, BPF, and BPAF, previous kinetic data from hepatic microsomes³² were used (Supporting Information Figures S3 and S4 and Supporting Information Table S9). Data generated from those various experiments was considered homogeneous since compositions of homologous UGTs are strongly correlated between commercially available liver microsomes and S9 fractions.⁸² Among BPA analogs, BPB was glucuronidated the most and BPS the least efficiently (Figure S3). Of note, BPA glucuronidation in human hepatic S9 fractions in the present study was faster than that previously reported in microsomes. To the best of our knowledge, glucuronidation kinetics for BPB, BPE, and BPM has not been previously reported. Intestinal glucuronidation of BPA, BPS, BPAF, and BPF was previously characterized in human intestinal microsomes and cell lines,^{32,83–85} even though bisphenols are metabolized predominantly in the liver.³² In line, we observed negligible glucuronidation rates in small intestinal S9 fractions only for BPA, BPB, BPE, and BPM. Intestinal glucuronidation could not be characterized for all bisphenol analogs and was not included in the model. Hepatic sulfation was also not represented due to the lack of comprehensive data for all analogs.⁸⁶ However, the latter might have a significant importance in children, particularly in girls,⁷⁰ which is unfortunately beyond the scope of the present model.

Rates of transport via efflux pumps were introduced as the parameter “EHC_r” and fitted by incrementally adjusting this

Table 2. Comparison of Measures and Predictions of Toxicokinetic Parameters for Oral Ingestion of BPA or BPS

	BPA	BPA glucuronide	BPS	BPS glucuronide
		$C_{\max}$ (nM)		
in vivo	0.43 (0.3–0.7) ^a	286 (173–386) ^a	8.79 (6.07–11.79) ^d	10.53 (7.77–14.2) ^d
present model ^b	0.27 (0.15–0.47)	228 (168–297)	14.1 (9.04–21.20)	29.2 (18.6–42.8)
previous model	1.31 (0.89–1.66) ^c	610 (580–648) ^c	10.1 (7.96–12.24) ^e	
		$t_{\max}$ (h)		
in vivo	1.6 (0.5–2.2) ^a	1.2 (0.8–2.2) ^a	0.5 (0.25–0.5) ^d	1.0 (1.0–1.0) ^d
present model ^b	0.70 (0.61–0.93)	1.16 (1.00–1.31)	2.15 (1.9–2.3)	1.0 (0.99–1.25)
previous model	0.72 (0.67–0.77) ^c	0.94 (0.93–0.97) ^c	0.57 (0.26–0.88) ^e	
		AUC (nM x day)		
in vivo	2.5 (1.4–5.7) ^a	680 (571–1'210) ^a	40 ^{31–49, d}	78.7 (54–103) ^d
present model ^b	3.97 (2.38–6.64)	1'036 (710–1'511)	163 (100–264)	191.3 (125–284)
previous model	4.15 (2.91–5.15) ^c	1'111 (1'050–1'210) ^c	39.8 ^{37–43, e}	

^aConcentrations measured in 10 men by Teeguarden et al.,³⁰ following administration of 30 $\mu\text{g/kg}$ bw d6-BPA in 355 mL of soup. ^bIntervals of variability of the present model predictions were set as the first and third quartiles of the MC simulations. ^cPreviously modeled by Karrer et al.³²

^dConcentrations measured after administration of 8.75 $\mu\text{g/kg}$ bw of d4-BPS in a chocolate cookie to 4 men and 3 women.³¹ ^ePreviously modeled by Oh et al.³¹

parameter in rat models until the predicted blood concentrations matched in vivo measurements at later time points. The resulting values were then directly used in human models (Table S10). Thereby, a similar high EHC_r rate was predicted for each bisphenol, with more than 63% of glucuronidated bisphenols leaving the liver through biliary excretion. Regarding sex differences, EHC_r values were slightly higher in females due to higher BPAF concentrations measured in female rats 24 h post administration.⁴⁸ Additionally, the half-life of BPF was longer in female rats compared to that in males.⁶³

Tissue-to-blood partition coefficients were estimated using a uniform approach based on an established QSAR method (Table S5).^{50,51} As discussed for a previous PBK model for some bisphenols,³² lipophilicity had an important influence on partition coefficient values. Overall, in a data-poor context, determining partition coefficients is largely hindered by the lack of tissue composition data available in the literature. Further characterization of sensitive endocrine tissues, such as the ovaries, breasts, and uterus, is needed to enhance the predictive scope of models used in female reproductive toxicity risk assessment.^{87,88}

The present PBK model incorporates 52 parameters, among which 14 depend directly or indirectly on logP and pK_a (namely, partition coefficients, P_{app}, and EHC_r). These physicochemical properties were predicted from molecular structures using two independent software tools (ChemDraw 20.0 and Chemaxon Playground v1.6.2.). Given the uncertainty inherent in such in silico predictions, we enhanced model robustness by averaging the values obtained from both methods (Tables S3 and S4). While this approach reduces potential biases, differences in physicochemical inputs can still influence PBK model outcomes, highlighting the importance of careful selection and validation of these parameters.

Finally, the multimodal parametrization strategy, while designed for broad applicability across structurally related compounds, also facilitates future refinement. As new experimental or computational data become available, individual parameters can be re-evaluated without altering the model's overall structure. For example, dermal exposure to bisphenols (which may be of equal or even higher toxicological relevance than dietary exposure)^{89–91} could not be included at this stage due to the lack of transferable and compound-

specific input data, but may be considered in future model expansions.

PBK Model Evaluation. The performance of the present model in predicting blood concentrations of BPA, BPS, and their metabolites was evaluated by using human experimental data. Moreover, the distribution of predicted concentrations obtained by resampling sensitive parameters in MC simulations, to account for parameter variability and uncertainty, was compared with the observed ranges of measured values for blood $C_{\max}$ (BPA, BPA glucuronide, BPS, and BPS glucuronide), $t_{\max}$ (BPA and BPA glucuronide), and AUC (BPA, BPA glucuronide, and BPS glucuronide).

Concerning BPA, the present model was evaluated with measures from Teeguarden et al.,³⁰ and predicted blood $C_{\max}$ values for BPA and its glucuronide were on average 63% and 80% of the observed values, respectively (Table 2). In the present predictions, $t_{\max}$ for BPA was lower than the observed value, while $t_{\max}$ for BPA glucuronide was similar. Predicted AUC for BPA and predicted AUC for BPA glucuronide were both about 1.5 times higher than those observed. Five of six predicted values were within the measurement ranges, and the range of predictions predicted by MC simulations overlapped with all measured ranges. These results indicate that the present model generates reliable predictions of the BPA concentrations.

The present model was also evaluated for BPS, using measures from Oh et al.³¹ Predicted blood $C_{\max}$ values for parent and glucuronide were on average 1.6 and 2.8 times higher than in vivo observed values, respectively. The average predicted $t_{\max}$ was delayed by 1.7 h for the parent compound and very similar to the measure for the metabolite. The mean predicted AUC value for BPS glucuronide was about 2.4 times higher than the in vivo value, while the mean predicted AUC for BPS parent was 4 times higher than observed.

Furthermore, the performance of the present model was compared to previous modeling efforts for BPA and BPS to assess the reliability of predictions using different parametrization approaches. The present model more accurately predicted internal bisphenol concentrations than the model developed by Karrer et al.³² for five out of six analyzed toxicokinetic parameters (average predicted values for $C_{\max}$, $t_{\max}$, and AUC for BPA and BPAG), suggesting superior performances of the present model for BPA (Table 2).^{31,32}

Although predictions for BPS toxicokinetics were less accurate than those from the model by Oh et al.,³¹ the latter was built by fitting parameters directly to human experimental data. While this ensures close reproduction of observed concentrations, it may lead to overfitting and limits the predictive value beyond the specific test conditions. In contrast, the present model was constructed from independent biological and chemical data, which offers predictive capabilities beyond the calibration context and supports extrapolation to untested scenarios. For both BPA and BPS, comparisons of predicted blood $C_{\max}$ values demonstrated that the present multimodal parametrization approach achieves equal or greater similarity to in vivo measures compared to in vivo data fitting methods used in previous models (Table 2). Moreover, since models built in data-poor situations cannot be calibrated, their predictions are limited by increased uncertainty. For instance, the PBK model established by Karrer et al.³² was calibrated with in vivo blood concentration values.³¹ An important aspect consists of testing the calibrated model against data that were not used to calibrate the model. Therefore, the present PBK model is the first to be evaluated for two different bisphenols and their metabolites, which ensures more reliable predictions.

The relative importance of excretion routes was evaluated through mass balance analysis (Table S11). Model predictions align closely with experimental data from Thayer et al.⁷² and Teeguarden et al.³⁰ Thayer et al.⁷² reported that 94% (range: 86–98%) of total BPA was excreted within 12 h post exposure, while our model predicts 97% and 96% urinary excretion in men and women, respectively. At 24 h, urinary excretion reached 98% (range: 92–100%) in Thayer et al.⁷² and 104% (range: 90–116%) in Teeguarden et al.;³⁰ our model estimates 99% for adult models of both sexes. By 48 h, Thayer et al.⁷² observed 80–110% cumulative urinary excretion, while the present model predicts 99%. Regarding unconjugated BPA, Teeguarden et al.³⁰ and Thayer et al.⁷² reported urinary fractions of 0.03% and 0.029%, respectively; the present model predicts around 0.01%. These results support the model's accuracy in reproducing urinary kinetics and excretion profiles.

Calculated renal and biliary clearance values indicate that CL_R was 10.1 L/h, while CL_{bile} was 17.0 L/h, yielding a CL_{bile}/CL_R ratio of 1.7. The corresponding bile-to-plasma concentration ratio was ~586. Such a prominent biliary route is physiologically plausible, as biliary clearances of 30 L/h or higher can be achieved, and bile-to-plasma ratios can approach 1000 for compounds that are strong substrates of apical efflux transporters.⁷³ High biliary clearance is particularly expected for glucuronides, which are polar and ionized ($pK_a \sim 3$) and have molecular weights of >350 g/mol,⁷³ properties shared by all metabolites considered in the present work. Additionally, the apparent fraction absorbed for BPA was 2.61, exceeding 1, which is consistent with extensive EHC. Nevertheless, it should be noted that EHC was scaled from rat in vivo data, which may not fully reflect human physiology. As such, while the predicted clearance values are physiologically plausible, it remains to be confirmed whether such high biliary clearance occurs in reality and whether this is specific to bisphenols.

Model evaluation also highlighted the need to improve the parametrization of ADME processes downstream of the systemic distribution. Indeed, all AUC values were overestimated compared to in vivo measures (Table 2). Overestimation of EHC is a primary hypothesis to explain the reduced predictive performance of the present model, especially for BPS, where we observe both an overestimation

of the AUC and a delayed $t_{\max}$. The EHC rate significantly impacted all of the concentrations analyzed with LSA (Figure S5), which aligns with its well-established influence on xenobiotic retention within organisms. However, it was not identified as an influential or interacting parameter in GSA (Figures S6 and S7). The motivation to model an EHC rate from rat toxicokinetic data measured in rats stemmed from the availability of respective blood concentration measurements beyond 24 h post administration and the confirmed existence of EHC for bisphenols in these animals.²⁷ However, interspecies differences in EHC have previously been observed between rodents and humans and may explain the suspected inaccuracies in the EHC modeling in the present model. Particularly, MWs of three out of seven bisphenol glucuronides are between the species' thresholds for excreting chemicals via bile (Table S4), which is estimated to be 400 Da for rats and 475 Da for humans.⁶⁴ Moreover, while the prolonged elimination time of BPS suggests it undergoes EHC in humans,³² this observation was not described for other bisphenols that might not even be substrates for human hepatic canalicular transporters.⁹² Ultimately, the EHC rates used in the present model were higher than those of previous estimations. In Yang et al.,²⁸ only ten percent of glucuronidated BPA underwent EHC, a rate seven times lower than our predictions (Table S10). Despite the perfectibility of EHC parametrization, we still considered the present approach to be acceptable as it resulted in AUC predictions that were closer to in vivo measurements than those obtained using lower EHC estimates (Table 2).³²

In the absence of tissue-specific toxicokinetic data in humans (which cannot ethically or practically be obtained under controlled exposure conditions), tissue concentration predictions generated by the present model were compared to available human biomonitoring data (Table S12).^{21,22,93–99} These comparisons aim to assess the general consistency of model predictions with reported concentrations of bisphenol analogs in various biological matrices and tissues while acknowledging the significant limitations inherent to such comparisons. Overall, model-predicted tissue concentrations were lower than human biomonitoring levels by approximately 2 orders of magnitude across blood and solid tissues, whereas urinary concentrations were overpredicted relative to population measurements. Together with the exploratory nature of our comparison, these discrepancies highlight gaps between our simulations and real-world exposures as well as the influence of interindividual variability. They also suggest that alternative parametrizations of EHC may be needed to better reproduce observed tissue concentrations.

A potential limitation of the PBK models is their incomplete representation of the human population. Despite efforts to broaden the applicability of the present models by considering physiological differences specifically for adult men and women and male children and toddlers, some population groups are not considered, e.g., elderly individuals, pregnant women, and those with liver or kidney impairments. While age-related scaling of UGT2B17 and MRP2 expression and MC simulations introduce variability, genetic polymorphisms and other sources of interindividual differences are not fully captured. Exposure was simulated via three daily meals using average EFSA doses over 96 h, which does not reflect lifelong, intergenerational, diverse environmental exposures, or potential mixture effects with other endocrine disruptors.¹⁰⁰ Geographic, socioeconomic, and occupational factors, as well

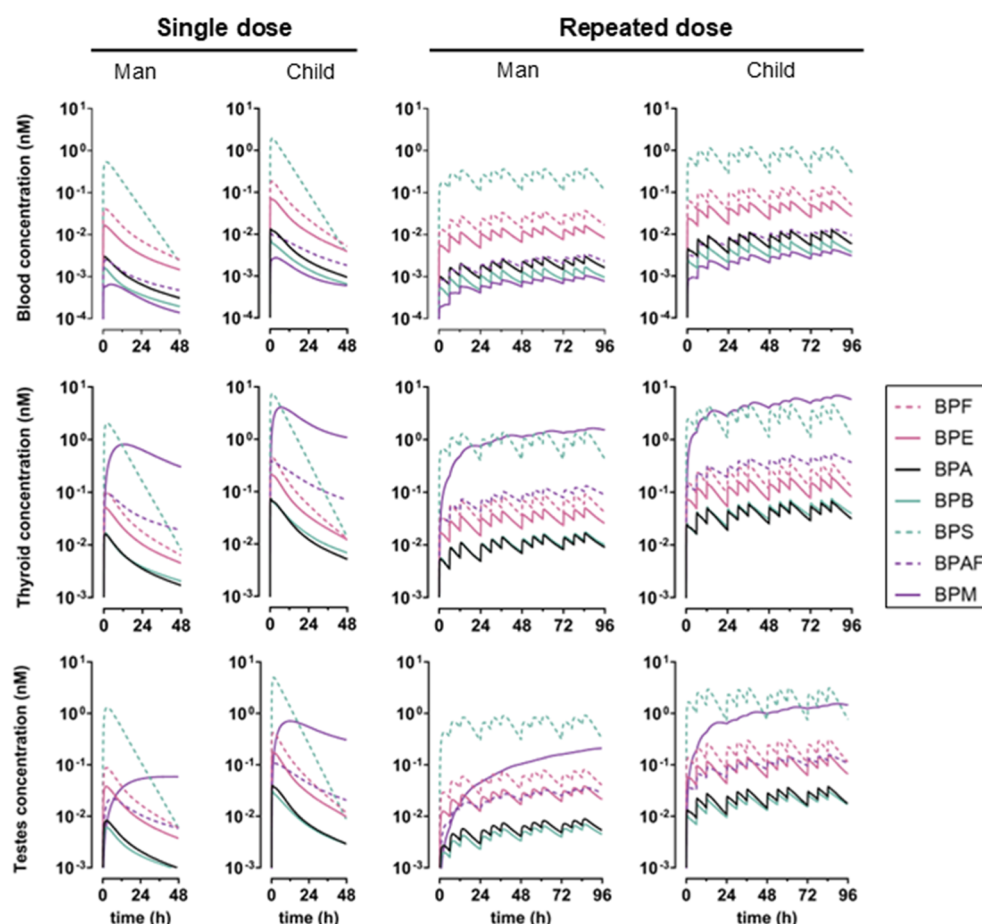


Figure 3. Predicted concentrations profiles (median of 10'000 MC simulations) of BPA, BPAF, BPB, BPE, BPF, BPM, and BPS in blood, thyroid, and testes in man and child models after single or repeated dose (exposure scenarios 5, 7, 9, and 11 of Table 1). Bisphenols are ordered by ascending MW. Since results are very similar between age groups, only man and child model results were represented for simplicity.

as vulnerable populations, were also not considered, meaning predictions represent average conditions rather than the full human population.

Sensitivity Analysis of Model Parameters. In LSA, seven parameters significantly influenced bisphenol concentrations across all key organs and human physiological models: body weight, cardiac output, ECHr, and hepatic glucuronidation kinetics ($V_{\max}$, K_m , liver volume, and liver blood flow; Figure S5). The scaling factor to account for age differences in glucuronidation kinetics (SFg) was also found sensitive in all non-adults analysis, supporting the use of in vitro over in silico metabolism parametrization for greater reliability. Frequently sensitive parameters also included the gastric emptying half time (GEst) and the blood debit to slowly perfused organs (QSL).

GSA confirmed that model uncertainty is primarily driven by the distribution of hepatic glucuronidation kinetics values. The Morris screening test yielded results consistent with the LSA, identifying $V_{\max}$, GEst, liver blood flow, and liver volume as influential parameters across all models, with SFg also being important in child and toddler models (Figure S6). These findings are consistent with the expected sensitivity due to the non-linearity of Michaelis–Menten kinetics. Further quantification of main and interaction effects using eFAST revealed that variance in model outputs is largely driven by interactions between parameters (Figure S7).

Internal Exposure Assessments. Concentrations of BPA, BPAF, BPB, BPE, BPF, BPM, and BPS were predicted in the blood, thyroid, testes, and breast tissues of man, woman, toddler, and child models following single or repeated environmentally relevant oral exposure. For all, higher concentrations were predicted in blood, thyroid, and testes in non-adult models compared to adult models (Figures 3 and S8 and Table 3), which aligns with previous predictions.³² Of note, internal maximal concentrations in children and toddlers are particularly increased due to the application of an age-related scaling factor (SFg) for UGT2B15 enzyme expression and correspondingly reduced glucuronidation.^{70,71} For repeated exposure, we predicted bisphenol accumulation in compartments and the attainment of the steady state over the course of 96 h, except for BPM and BPAF in the thyroid, testes, and breasts (Figures 3, 4, and S8 and Table 3).

The toxicokinetics of BPE, BPB, and BPM were modeled in humans for the first time and compared to other analogs. BPS, BPF, and BPE had the highest blood concentrations, were among the highest in tissues, and reached the steady state within 24–48 h (Figures 3, 4, and S8 and Table 3). BPE, with the third slowest glucuronidation rate and third lowest partition coefficients (Figure S3 and Table S5), consistently exhibited high concentrations across most organs (Figures 3, 4, and S8). In contrast, BPB had faster glucuronidation rates and higher blood–tissue partition coefficients (Figure S3 and Table S5), leading to its longer retention in the liver, where it was

Table 3. Highest $C_{\max}$ Values per Bisphenol and per Key Organ after Repeated Dose (Exposure Scenarios 9 to 12), among the Human Models^a

bisphenol	max. $C_{\max}$ blood (pM)	max. $C_{\max}$ breasts (pM)	max. $C_{\max}$ testes (pM)	max. $C_{\max}$ thyroid (pM)
BPS	1221 ^{b,c}	149 ^d	3120 ^{b,e}	4658 ^c
BPF	140 ^c	163 ^d	313 ^c	351 ^c
BPE	62 ^e	133 ^d	154 ^e	192 ^e
BPAF	14 ^g	244 ^{d,f}	157 ^{e,f}	547 ^{e,f}
BPA	13 ^g	45 ^d	38 ^e	68 ^e
BPB	7 ^g	45 ^d	32 ^e	76 ^e
BPM	4 ^g	335 ^{b,d,f}	1536 ^{e,f}	6878 ^{b,e,f}

^aBisphenols are ordered by descending blood concentration. ^bHighest concentration per compartment (among all bisphenols). ^cConcentration predicted by the toddler model. ^dConcentration predicted by the woman model. ^eConcentration predicted by the child model. ^fConcentration for which the steady state was not reached. ^gConcentration predicted by the toddler and child model (with difference <1 pM).

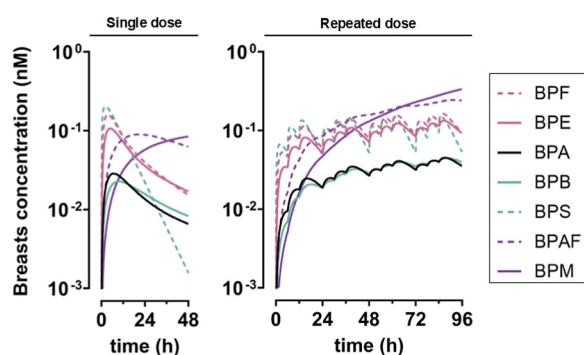


Figure 4. Predicted concentrations profiles (median of 10'000 MC simulations) of BPA, BPAF, BPB, BPE, BPF, BPM, and BPS in breasts after single dose exposure after single or repeated dose (exposure scenarios 6 and 10 of Table 1). Bisphenols are ordered by ascending MW.

rapidly metabolized and excreted. BPB and BPA shared similar parameter values, resulting in lowest tissue concentrations, particularly in the thyroid, testes, and breasts, and steady-state achievement at ~48 h (Figures 3, 4, and S8). More efficient glucuronidation rates in the current model predicted lower BPA concentrations compared to the previous model.³² BPM and BPAF are not structurally similar, but concentrations of both were among the lowest in blood and among the highest in thyroid and breasts (Figures 3, 4, and S8). While BPM underwent rapid hepatic glucuronidation (Figure S3), its kinetics differed significantly due to tissue/blood partitioning being several degrees of magnitude higher than that of other bisphenols. As a consequence, BPM concentration did not decrease during the 48 h following single dose exposure, did not reach the steady state in thyroid, testes, and breasts, and persisted beyond 96 h of simulation (Table 3 and Figures 3, 4, and S8).²²

The present models predict bisphenol concentrations in three toxicologically relevant compartments related to evident toxic properties of BPA: thyroid, breasts (in women), and testes (in males). It is the first to predict concentrations of BPA analogs in the thyroid, for which their hormone-disrupting potential was being previously evaluated in vitro at much higher doses.¹⁰¹ Further, our model is the first to explore internal exposure to BPA alternatives in breasts,

despite the carcinogenic potential of xenoestrogens in breast tissue. The potential effects of BPB, BPA, BPF, and BPS on estrogen receptor α , which can lead to tumor progression and metastasis, were previously studied in cell lines MCF-7, MDA-MB-231, and T47D cells at concentrations ranging from 10^{-11} to 10^{-4} M in the absence of precise exposure data.¹⁰² Concerningly, BPA was reported to exert increased combinatory estrogenicity in coexposure with other xenoestrogens.^{6,103} BPA analogs are commonly observed simultaneously in human biomonitoring.^{16,18–20,22–25} In this light, PBK models could serve as a starting point of selecting concentrations of BPA analogs to test for mixture effects and combinatory toxicity.

Compared to the previous model by Karrer et al.,³² these models more precisely parametrized male testes physiology⁵³ and included three other bisphenols of potential reproductive toxicity. For instance, the impact of BPA, BPF, BPS, BPE, and BPB on the endocrine function of adult human testis explant cultures was tested at concentrations between 10^{-9} and 10^{-5} M.¹⁰⁴ Cytotoxicity and phenotypic marker observations due to BPA, BPAF, BPF, BPS, and BPM exposure in germ and steroidogenic cell lines were also performed at concentrations ranging from 1 nM to 100 μ M using single-cell high-content imaging.¹⁰⁵ Our predictions provide a toxicokinetic perspective to these studies, clarifying their results and supporting further studies of BPA analog toxicity.

The presented PBK models predict significant differences in internal concentrations following oral exposure to equal levels of BPA and six analogs, highlighting the critical role of toxicokinetics in individual chemical toxicity. This underscores the need to avoid extrapolating BPA toxicity to structurally similar chemicals¹⁰⁶ and emphasizes the urgency of conducting chemical-specific risk assessments for widely used BPA alternatives.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.est.5c00513>.

Examples of biomonitoring data indicating the importance of human exposure to bisphenol analogs; workflow of PBK model development, validation, reporting and dissemination; physiological parameters; physico-chemical properties of bisphenols included in the present study; physicochemical properties of bisphenol glucuronides; partition coefficients (P-) and unbound fractions (fu, FU-); chemicals, reagents, and enzymes used for glucuronidation kinetics; HPLC retention times of bisphenols; parameter variability estimates; estimated Papp across the Caco-2 monolayer determined by various techniques; predicted rates of glucuronidation of bisphenols in the human adult liver at physiologically relevant concentrations; hepatic glucuronidation kinetics; kinetics parameters for hepatic glucuronidation; EHC rate values; frequency of sensitivity of parameters across sensitivity analysis; Morris screening exercise for AUC of the blood concentration of BPA; Lowry plots of the eFAST quantitative measures for AUC of BPA blood concentration; mass balance analysis of excretion; predicted concentration profiles in woman (blood and thyroid) and toddler (blood, thyroid, and testes) models after single or repeated dose; and biomonitoring comparison (PDF)

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Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

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